

Question 1

a) data = [red, blue, blue, red, green, grey] $n=6$ 4 category

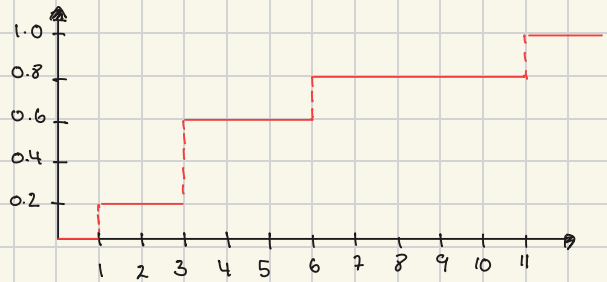
$$X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b) $\hat{p}_{\text{red}} = \frac{2}{6}$ $\hat{p}_{\text{blue}} = \frac{2}{6}$ $\hat{p}_{\text{green}} = \frac{1}{6}$ $\hat{p}_{\text{grey}} = \frac{1}{6}$

$$\hat{p} = \left[\frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{6} \quad \frac{1}{6} \right]$$

c) $X = [1, 3, 3, 6, 11]$ $n=5$

Threshold	ECDF
$t < 1$	0
$1 \leq t < 3$	$\frac{1}{5}$
$3 \leq t < 6$	$\frac{3}{5}$
$6 \leq t < 11$	$\frac{4}{5}$
$t \geq 11$	$\frac{5}{5}$



Question 2

a) $y = a + bx$ $b > 0$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(a + bx \leq y) \\ &= P(bx \leq y - a) \\ &= P\left(x \leq \frac{y - a}{b}\right) \\ &= F_X\left(\frac{y - a}{b}\right) \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) \\ &= \frac{d}{dy} F_X\left(\frac{y - a}{b}\right) \\ &= \frac{1}{b} f_X\left(\frac{y - a}{b}\right) \end{aligned}$$

b) $X = [1, 3, 4]$

$$\bar{x} = \frac{1 + 3 + 4}{3} = \frac{8}{3}$$

$$c) \frac{1^2 + 3^2 + 4^2}{3} = x^2$$

$$x^2 = \frac{1+9+16}{3} = \frac{26}{3}$$

$$d) \left(\frac{8}{3}\right)^2 = \frac{64}{9} \quad \text{No}$$

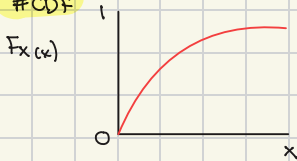
Question 3

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$

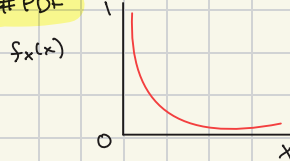
$$a) f_x(x) = \frac{d}{dx} F_x(x) \quad \downarrow$$

$$f_x(x) = \begin{cases} 0 & x < 0 \\ e^{-x} & x \geq 0 \end{cases}$$

#CDF



#PDF



$$b) F_x(x) = 1 - e^{-x} \quad x \geq 0$$

$$P(x \geq 3) = 1 - P(x \leq 3)$$

$$= 1 - F_x(3)$$

$$= 1 - (1 - e^{-3})$$

$$= e^{-3}$$

$$P(x \leq 35) = F_x(35)$$

$$= 1 - e^{-35}$$

$$c) F_Y(y) = P(Y \leq y)$$

$$= P(1 + 3X \leq y)$$

$$= P(X \leq \frac{y-1}{3})$$

$$= F_x\left(\frac{y-1}{3}\right) \Rightarrow F_x(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x} & x \geq 0 \end{cases} \Rightarrow F_Y(y) = \begin{cases} 0 & y < 1 \\ 1 - e^{-(y-1)/3} & y \geq 1 \end{cases}$$

#CDF



$$\begin{aligned}
 P(Y < 0) &= P(43X < 10) \\
 &= P(3X < 9) \\
 &= P(X < 3) \\
 &= F_X(3) \\
 &= 1 - e^{-3}
 \end{aligned}$$

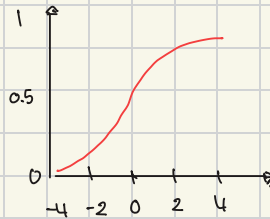
Question 4

$$F(x) = \frac{1}{1 + e^{-x}}$$

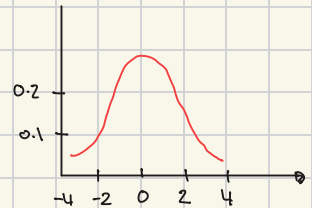
a) $f_X(x) = \frac{d}{dx} F_X(x)$

$$\begin{aligned}
 f_X(x) &= \frac{d}{dx} (1 + e^{-x})^{-1} \\
 &= -(1 + e^{-x})^{-2} (-e^{-x}) \\
 &= \frac{e^{-x}}{(1 + e^{-x})^2}
 \end{aligned}$$

CDF



PDF



b) $F_X(x) = \frac{1}{1 + e^{-x}}$

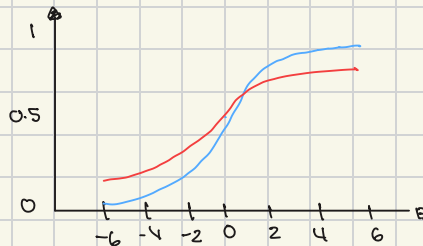
$$\begin{aligned}
 P(X > 0.8) &= 1 - F_X(0.8) \\
 &= 1 - \frac{1}{1 + e^{-0.8}} \\
 &= 0.3100
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq 0.3) &= F_X(0.3) \\
 &= \frac{1}{1 + e^{-0.3}} \\
 &= 0.5744
 \end{aligned}$$

c) $F_Y(y) = P(Y \leq y)$

$$\begin{aligned}
 &= P(2X - 1 \leq y) \\
 &= P(X \leq \frac{y+1}{2}) \\
 &= F_X\left(\frac{y+1}{2}\right)
 \end{aligned}$$

$$F_Y(y) = \frac{1}{1 + e^{-(y+1)/2}}$$



$$\begin{aligned}
 P(Y < 0) &= P(2X - 1 < 0) \\
 &= P(X < \frac{1}{2}) \\
 &= F_X(0.5) \\
 &= \frac{1}{1 + e^{-0.5}} \\
 &= 0.6275
 \end{aligned}$$

Question 5

a) $F(\text{median}) = \frac{1}{2}$ $F_X(x) = 1 - e^{-x}$

$$1 - e^{-m} = \frac{1}{2}$$

$$e^{-m} = \frac{1}{2}$$

$$-m = \ln\left(\frac{1}{2}\right)$$

$$m = -\ln\left(\frac{1}{2}\right)$$

$$= \ln(2)$$

b) $F(\text{median}) = \frac{1}{2}$ $F_X(x) = \frac{1}{1 + e^{-x}}$

$$\frac{1}{1 + e^{-m}} = \frac{1}{2}$$

$$1 + e^{-m} = 2$$

$$e^{-m} = 1$$

$$-m = \ln(1)$$

$$m = 0$$

c) $u = 1 - e^{-x}$

$$e^{-x} = 1 - u$$

$$-x = \ln(1 - u)$$

$$x = -\ln(1 - u)$$

$$0 < u < 1$$

d) $u = \frac{1}{1 + e^{-x}}$

$$u(1 + e^{-x}) = 1$$

$$1 + e^{-x} = \frac{1}{u}$$

$$e^{-x} = \frac{1 - u}{u}$$

$$-x = \ln\left(\frac{1 - u}{u}\right)$$

$$x = \ln\left(\frac{u}{1 - u}\right)$$

$$0 < u < 1$$